SAM Enhanced Semantic Segmentation for Remote Sensing Imagery Without Additional Training

Yang Qiao, Bo Zhong, Bailin Du, He Cai, Jinxiong Jiang, Qinhuo Liu, Aixia Yang, Junjun Wu, and Xiaoya Wang

***Abstract*— Semantic segmentation is a critical process in remote sensing image analysis, supporting various applications. The recent development of the segment anything model (SAM), a visual foundation model designed to segment-anything, highlights the potential of foundational models in computer vision. However, SAM generates segmentation results without category labels, and predictions from semantic segmentation models for remote sensing often exhibit excessive fragmentation and imprecise boundaries. To address these limitations, we propose a strategy that integrates SAM with semantic segmentation models, replacing the imprecise boundaries of remote sensing segmentation masks with the more boundary-accurate SAM masks while retaining the original semantic information. Subsequently, a framework is designed and realized to enhance the prediction results of semantic segmentation models for remote sensing imagery by leveraging the raw outputs generated by SAM. This approach requires no additional training, modification to the semantic segmentation model, or changes to the visual foundation model, making it efficient and straightforward compared with other methods. Specifically, experimental results on two well-known datasets, LoveDA Urban and ISPRS Potsdam, demonstrate the effectiveness and broad applicability of our approach. In addition, incorporating recent visual foundation models, such as SAM-HQ and semantic SAM, further improves segmentation accuracy. As these models advance, the potential of our framework to enhance the performance of semantic segmentation for remote sensing imagery will grow. The source code for this work will be accessible at https://github.com/qycools/SESSRS.**

***Index Terms*— Remote sensing, segment anything model (SAM), semantic segmentation.**

Received 22 October 2024; revised 18 December 2024; accepted 14 January 2025. Date of publication 29 January 2025; date of current version 21 February 2025. This work was supported in part by the National Key Research and Development Program under Grant 2021YFE0117400 and Grant 2022YFF1300200, and in part by the Natural Science Foundation of Hainan Province under Grant 422QN350. *(Corresponding author: Bo Zhong.)*

Yang Qiao, Bailin Du, and Xiaoya Wang are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: qiaoyang22@mails.ucas.ac.cn; dubailin20@mails.ucas.edu.cn; wangxiaoya23@mails.ucas.ac.cn).

Bo Zhong, Qinhuo Liu, Aixia Yang, and Junjun Wu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhongbo@aircas.ac.cn; liuqh@aircas.ac.cn; yangax@aircas.ac.cn; wujj@aircas.ac.cn).

He Cai is with the School of Computer Science and Engineering, Northeastern University, Shenyang 110169, China (e-mail: 1910610@stu.neu.edu.cn).

Jinxiong Jiang is with the Space Star Technology Company Ltd., Beijing 100086, China (e-mail: jiangjx503@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3531930

I. Introduction

SEMANTIC image segmentation is a fundamental task in computer vision, where each pixel is assigned a label, also known as pixel-level classification. With the continuous advancement of semantic segmentation technologies, they have been increasingly applied to address the inherently diverse and data-rich challenges in remote sensing [1], including environmental monitoring [2], land resource management [3], and disaster management [4]. In recent years, to address the unique challenges of remote sensing images, such as large size, complex backgrounds, and diverse categories, researchers have developed numerous specialized semantic segmentation models [5], such as the attentive bilateral contextual network (ABCNet) [6] and a UNet-like transformer (UNetformer) [7], to enhance the accuracy of semantic segmentation in RS images. However, the diversity of RS images poses significant challenges to the generalization ability of these models [8].

Inspired by large language models, many researchers are developing large-scale visual foundation models to learn general purpose representations for computer vision tasks. Notable examples include CLIP [9] and DALL-E [10]. These models are designed for tasks such as image generation and are trained on extensive datasets to capture complex features, patterns, and representations in visual data [11]. The segment anything model (SAM), recently proposed by Kirillov et al. [12], enables segmentation within an image by providing visual prompts such as bounding boxes, points, or masks to specify the content to be segmented. Trained on over 1.1 billion masks from 11 million images, SAM possesses a robust zero-shot learning capability, making it highly effective for downstream visual tasks. The advent of SAM has significantly advanced the development of foundational models in computer vision. Consequently, numerous efforts have emerged to explore improvements to SAM [13], [14], [15], [16] and its application to various tasks, including image and video processing [17], [18], [19]. However, due to the absence of semantic labels in the masks generated by SAM and the inherent differences between natural and remote sensing images, SAM's performance in remote sensing semantic segmentation has been less than ideal [20].

To address the limitations of SAM on remote sensing images, researchers have recently explored various methods to enhance SAM's performance in remote sensing image tasks. These approaches can be broadly categorized into two types.

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

1) Generate SAM output by the default SAM: directly use the SAM output as the segmentation result for a downstream task [21], augment the segmentation result with the SAM output [22], [23], or combine the SAM output with few-shot or zero-shot learning strategies [24], [25], [26]. For instance, [22] leverages SAM's ability to automatically delineate cropland units from high-resolution satellite imagery without additional training, thereby significantly improving the quality of raw CDL data for major U.S. agricultural regions. RS-CLIP [25] uses a curriculum learning strategy involving multiple stages of model fine-tuning to enhance the performance of zero-shot remote sensing scene classification.
2) Applying prompt learning techniques to adapt SAM for the remote sensing domain, thereby generating more accurate outputs [17], [18], [27], [28], [29]. For example, RSPrompter [17] focuses on learning how to generate prompt inputs for SAM, enabling it to automatically acquire instance-level semantic masks and, thus, enhancing the capabilities of the SAM framework. While these advancements have significantly expanded SAM's application in remote sensing, they often require the artificial design of complex fine-tuning mechanisms or prompt learning strategies, as well as retraining of the remote sensing semantic segmentation model or the visual foundation models.

To cope with the aforementioned challenges, we developed a simple yet effective framework for postprocessing semantic segmentation results using the raw outputs of SAM. Our observations indicate that although SAM may not accurately segment every part of a remote sensing image due to differences between remote sensing and natural images, some interoperability between them still exists. By retaining the segments where SAM performs well, we can leverage the useful information SAM learns from natural images to improve remote sensing image segmentation. Therefore, we propose a straightforward framework that accurately retains SAM's boundary recognition on remote sensing images. By combining these with the semantic information generated by the semantic segmentation model, we can obtain high-quality masks with both precise boundaries and semantic information. Our approach directly uses SAM's raw outputs to enhance the semantic segmentation model's predictions, effectively leveraging SAM's potential in remote sensing. The contributions of this work are summarized as follows.

1) A simple postprocessing approach is proposed to enhance the predictions of the remote sensing semantic segmentation model by directly using the raw outputs of SAM. Notably, in contrast to the existing SAM-based methods in remote sensing, our approach does not require specific training strategies, prompt learning techniques, or pseudolabel generation.
2) We propose a novel framework to retain the masks with accurate boundary from SAM and combine them with the masks containing semantic information generated by the semantic segmentation model to produce high-quality masks with both precise boundaries and semantic information. Since it does not introduce additional modules into the training process, it can be processed directly on the baseline model's prediction results without introducing additional training.
3) Extensive experiments on two well-known publicly available remote sensing datasets, LoveDA Urban and ISPRS Potsdam, using eight representative semantic segmentation models from four different types and three visual foundation models, confirm that the proposed method can be widely adapted to semantic segmentation tasks across different datasets and models. We believe that this simple and efficient method can advance the application of visual foundation models, such as SAM, in the semantic segmentation of remote sensing images.

The remainder of this article is organized as follows. Section II first briefly introduces the development of SAM and the application of SAM in remote sensing field. Then, we describe our proposed approach in detail in Section III. Section IV presents both the quantitative and qualitative results, which demonstrate the effectiveness of our method. Finally, the conclusion is given in Section V.

II. Related Works

A. Semantic Segmentation in Remote Sensing

The rapid development of deep semantic segmentation techniques has led to the widespread adoption of various image segmentation architectures in remote sensing images, achieving remarkable performance. However, remote sensing images present unique challenges compared with natural scenes, including large size, complex backgrounds, significant scale variations, dense object arrangements, and low spatial resolution. These factors pose substantial scientific challenges for semantic segmentation in remote sensing imagery [5].

To address these issues, researchers have designed specialized architectures for semantic segmentation. For instance, Yu and Koltun [30] proposed the use of dilated convolutions to effectively enlarge receptive fields without increasing parameters, thereby capturing global context. Ronneberger et al. [31] and Chen et al. [32], [33] used multiscale context modeling, where low-level feature information is preserved to maintain high-quality spatial details, while high-level semantic information is incorporated to improve segmentation accuracy. Given the high computational cost of extracting global context information and processing large-scale feature maps, a series of lightweight networks have been developed to accelerate computation while balancing accuracy and efficiency [6], [34]. To adaptively extract features closely related to specific tasks, various attention mechanisms have been widely used in semantic segmentation. These mechanisms primarily focus on integrating attention with deep learning models, such as convolutional neural networks (CNNs) [35], [36], or adopting pure attention-based models, primarily transformers [37], [38]. These approaches can be broadly categorized into the following four groups: CNN-based lightweight networks, such as ABCNet [6] and ShelfNet [34], CNN-based attentional networks, such as FANet [35] and MANet [36], hybrid transformer-based networks with a transformer-based encoder and a CNN-based decoder, such as DC-Swin [39] and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

UNetFormer [7], and fully transformer-based networks such as SegFormer [37] and FT-UNetFormer [7].

B. Segment Anything Model

Driven by advances in deep learning, tasks such as semantic segmentation, instance segmentation, and panoptic segmentation have made significant progress. However, existing segmentation models often require specialized training and fine-tuning for specific tasks or datasets, making cross-domain generalization challenging. To address this issue, Meta AI introduced the first visual foundation model, the SAM, with the goal of "segmenting anything" by simplifying the interaction process [12].

SAM's architecture comprises three main components: an image encoder, a prompt encoder, and a mask decoder. The image encoder, based on the Vision Transformer (ViT) [40] architecture, extracts global features from the input image. The prompt encoder converts user-provided prompts (such as clicks, bounding boxes, or text descriptions) into prompt vectors aligned with the image features. The mask decoder then combines the image features from the image encoder and the prompt vectors from the prompt encoder to generate the final segmentation mask. SAM's design allows it to perform exceptionally well across various application scenarios without the need for specialized training or fine-tuning.

SAM's generality and flexible prompting mechanism enable it to adapt to a wide range of image segmentation tasks across various domains. Recently, with SAM's significant progress in many tasks across different fields, researchers have further optimized the model. For example, FastSAM [13] and MobileSAM [14] greatly improve SAM's inference speed, while SAM-HQ [15] and Semantic-SAM [16] primarily enhance its inference accuracy. These technical advancements further promote the application and development of SAM across various fields.

C. SAM in Remote Sensing

SAM's training relies on a large-scale dataset called SA-1B, which contains over a billion annotated image regions. However, these data are derived exclusively from natural imagery, and the differences between natural and remote sensing images primarily stem from factors such as acquisition methods, spectral and spatial resolution, object scale and coverage, content complexity, and limited annotation data [29], [41], [42]. RingMoSAM [29] addresses this issue by constructing a large-scale dataset with millions of segmentation instances collected from multiple open-source datasets in the remote sensing field to train a foundational model for multimodal remote sensing image segmentation. RSPrompter [17] introduces a novel prompt learning technique that enhances SAM's capability to facilitate instance segmentation in remote sensing images. SSRS [23], building on the original cross-entropy loss, introduces object consistency loss and boundary preservation loss by leveraging SAM-generated objects (SGOs) and SAM-generated boundaries (SGBs) to enhance model training. In addition, there are several strategies [25], [26] based on vision–language models to address the challenging tasks of few-shot and zero-shot classification in remote sensing, although certain limitations remain to be addressed [24]. These advancements have significantly enhanced the applicability of SAM in the remote sensing domain; However, they still require retraining or fine-tuning of visual foundation models or semantic segmentation models, which inevitably limits the further development of SAM in remote sensing. In light of this, it is imperative to design a method that can directly use SAM's raw outputs to enhance the performance of remote sensing semantic segmentation models.

III. Methodology

In this study, we refer to each segmentation mask as an object, treating it as an individual enclosed region. Since each segmentation mask can be considered an object with a distinct enclosed region, we define the segmentation masks as objects. We specifically call SAM masks "SAM objects" and RS semantic segmentation masks "RS objects." Each object has two attributes: shape and area, while the RS object has an additional semantic attribute that distinguishes it from the SAM object. We convert the segmentation masks of all individually closed regions into independent objects.

The schematic of the proposed framework is depicted in Fig. 1. In Fig. 1(a), we input the remote sensing images and use the prediction results generated by the semantic segmentation model to create an RS object list P, while the masks automatically generated by SAM are used to create an SAM object list S. In Fig. 1(b), we enhance the remote sensing semantic segmentation results using the output from SAM to enhance overall segmentation performance. Specifically, we design conditional statement (2), (3) to assign each object in the SAM object list S to a unique category c, generating the assigned object list S^A. Then, using condition (2), (3), the more accurate boundary of S^A replaces the mask of P, while preserving the semantic information from P. In Fig. 1(c), we address the issues that arise during the reconversion from a list of object types to a semantic segmentation map. In this section, we will provide a detailed explanation of *generate RS object and SAM object from mask*, *mask assignment and mask selection*, *adapting to remote sensing images* and *restore RS semantic segmentation image*.

A. Generate RS Object and SAM Object From Mask

In this work, we use a classical encoder–decoder network [6], [7], [34], [35], [36], [37], [39] as the baseline semantic segmentation model within the proposed framework. Given an input remote sensing image X, where $X \in \mathbb{R}^{H\times W\times 3}$, the semantic segmentation model produces a predicted segmentation output P^I, denoted as $P^I \in \mathbb{R}^{H\times W\times C}$, with C representing the number of land-cover classes. Using the semantic segmentation model, we obtain the initial semantic segmentation image P^I as illustrated in Fig. 1(a). All the objects with the same semantic label c in the semantic segmentation image P^I are stored in the same object list P_c, $P_c = [P_{c_1}, \ldots, P_{c_n}]$, P is the ensemble of P_c, $P = [P_1, \ldots, P_C]$, n is the number of objects with semantic label c, and $c \in [1, C]$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

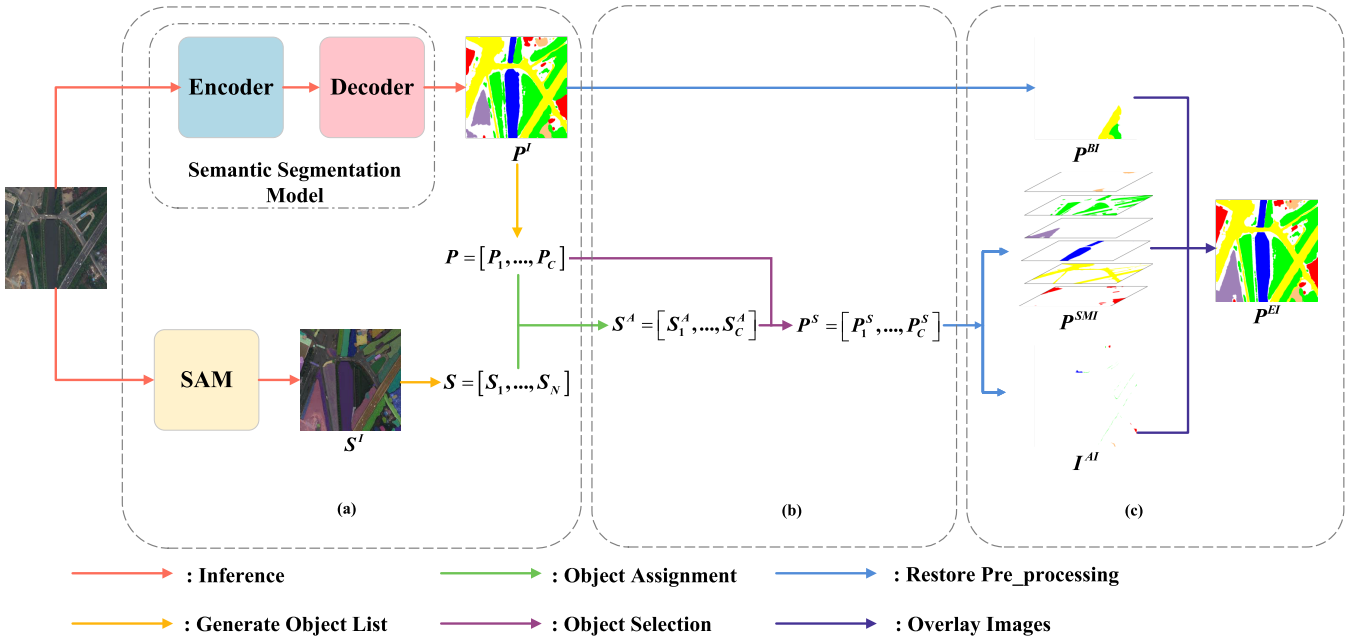


Fig. 1. Framework of our method. (a) Generate RS object and SAM object from mask. (b) Object assignment and object selection. (c) Restore RS semantic segmentation image. Superscript meanings: I: image, A: assigned, B: based, S: selected, M: merged.

The SAM provides grid-based prompting technology to automatically process images. Given an input remote sensing image $X \in R^{H\times W\times 3}$, the SAM is capable of generating segmentation masks S^I [12], at all possible locations specified in the grid prompt settings, as shown in Fig. 1(a). As the generated masks from the SAM lack semantic information, all the unannotated objects produced by the SAM are stored in a list, denoted as $S = [S_1, \ldots, S_N]$, where N represents the number of masks generated by SAM.

B. Object Assignment and Object Selection

Although remote sensing semantic segmentation models are capable of distinguishing various land-cover classes and discerning their locations within regions, their ability to precisely recognize object boundaries remains limited. In contrast, SAM excels in finely segmenting a large number of objects [12]. By integrating this capability, we propose a method that enhances remote sensing segmentation results using SAM, leveraging SAM's potential to refine object boundaries in segmentation outputs. As illustrated in Fig. 1(b), our method takes the remote sensing object list P and SAM object list S as inputs and returns the enhanced RS object list $P^S = [P_1^S, \ldots, P_C^S]$, and it undergoes two stages: object assignment and object selection. During the object assignment stage, we identify the category c with the largest area in each S_n based on the semantic segmentation image P^I and assign S_n to category c, as shown in (1), to obtain the SAM object allocation list S^A, $S^A = [S_1^A, \ldots, S_C^A]$, and $S_c^A = [S_{c_1}^A, \ldots, S_{c_n}^A]$, where n denotes the number of SAM objects labeled as c

$$c^* = \arg\max_c (S_n). \tag{1}$$

The ambiguity in semantic boundaries often arises from oversegmentation or undersegmentation [43], [44]. Our object selection strategy directly addresses these issues by selecting appropriate SAM objects based on the significant overlap between SAM objects and RS objects. When an RS object is merged with the surrounding background or parts of other objects, oversegmentation occurs. Therefore, if an RS object covers a large portion of a SAM object, it indicates that the SAM object may precisely represent the target object. Conversely, when an RS object covers only the most distinguishable part of the target object rather than the entire object, undersegmentation occurs. Thus, if an SAM object covers a substantial part of an RS object, it suggests that the SAM object may represent the complete target object.

Based on the intuitions mentioned above, we iterate through every SAM object assigned to category c. For each assigned SAM object S_a in the SAM object assignment list S_c^A, we compute o_a, the proportion of assigned SAM object S_a overlapped by RS object S_p, and o_p, the proportion of RS object S_p overlapped by assigned SAM object S_a

$$o_a = \frac{\text{Intersect}(S_a, S_p)}{S_a} \tag{2}$$

$$o_p = \frac{\text{Intersect}(S_a, S_p)}{S_p}. \tag{3}$$

As shown in Fig. 2, if the following conditions are met, the shape of the RS object S_p will be replaced by the shape of the assigned SAM object S_a, while preserving the semantic information of S_p.

1) $o_a > t_1$, indicating that at least t_1 of the SAM object's area is covered by the RS object, its main aim is to resolve the problem of undersegmentation, and it will enlarge the area of RS object.
2) $o_p > t_2$, indicating that at least t_2 of the RS object's area is covered by the SAM object, its main aim is to resolve

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

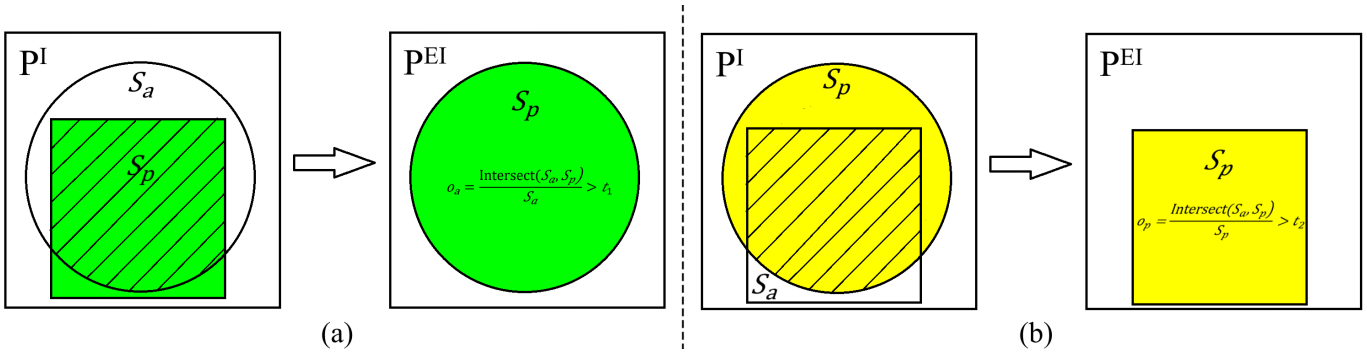


Fig. 2. Schematic of semantic segmentation image P^I processing to enhanced semantic segmentation image P^{EI}. The shaded area represents the intersection area. (a) $o_a > t_1$, its main aim is to resolve the problem of undersegmentation by enlarging the area of RS object. (b) $o_p > t_2$, its main aim is to resolve the problem of oversegmentation by reducing the area of RS object. The superscript E indicates the enhanced result.

the problem of oversegmentation, and it will reduce the area of RS object.

C. Adapting to Remote Sensing Images

Our object assignment and selection algorithm is inspired by [45]. While SAM generally performs well on natural images, it encounters challenges in specific domains, such as medical imaging or remote sensing images, where training datasets lack representation of these data types [46]. This results in suboptimal SAM performance due to two main factors.

1) The effectiveness of SAM segmentation depends on the contrast between the target object and its surrounding environment. When the contrast is high, SAM achieves excellent segmentation results. However, in remote sensing, as objects increasingly blend with their background, and their boundaries become indistinct or irregular, SAM's performance deteriorates. Therefore, we assign different thresholds t_1^c and t_2^c for different land-cover categories

$$t_1 \to t_1^c, \quad t_2 \to t_2^c. \tag{4}$$

2) There is a difference between the semantic segmentation model and the SAM model in perceiving the size of ground objects in remote sensing images. Sometimes, SAM oversegments a ground object by interpreting it as multiple distinct objects, while at other times, it predicts multiple ground objects as a single object due to unclear boundaries. These situations can lead to significant differences in the areas of objects segmented by SAM compared with those predicted by the semantic segmentation model. The above approach is effective when the area difference between the SAM mask S_n and the semantic mask S_p is small. When there is an overlapping area between the SAM assignment mask S_a and the semantic mask S_p, and $\text{area}(S_a) \gg \text{area}(S_p)$, then $o_p \to 1$. Conversely, $o_a \to 1$, it may lead to serious errors, as illustrated in Fig. 3. Based on this, we have modified (1) to better accommodate the remote sensing scenarios. We revised the SAM object assignment rule, as shown in (5) below. By constraining S_n with the threshold t_1^c, this equation prevents the occurrence of errors where $\text{area}(S_a) \gg \text{area}(S_p)$

$$c^* = \arg\max_c(S_n) \to c^* = \frac{S_n\left(S_n == \arg\max(S_n)_c\right)}{S_n} > t_1^c \tag{5}$$

where c^* represents various object types.

In the object selection phase, we added additional decision (6). When the temporary list tmp contains only one S_a, we limit its area to ensure that the difference with S_p is minimal. When tmp list contains multiple S_a, we merge them before comparing with (7), which helps avoid errors caused by $\text{area}(S_p) \gg \text{area}(S_a)$

$$\text{number}_{\text{tmp}} == 1 \text{ and } t_2^c * S_p < \text{tmp}_1 < \frac{S_p}{t_1^c} \tag{6}$$

$$\frac{\text{Intersect}(\text{Merge}(\text{tmp}), S_p)}{S_p} > t_2^c \tag{7}$$

where tmp refers to the list objects in S_a that satisfy (6) and (7) for temporary storage. The above algorithm is summarized in Algorithm 1.

D. Restore Semantic Segmentation Image

The method in [45] focuses solely on enhancing semantic segmentation results for localized foreground objects. When the SAM object S_n replaces the object S_p, changes in the area of S_p only impact the background category. In contrast, our method addresses the multicategory global semantic segmentation problem, which requires a global semantic image restoration operation. For datasets like Potsdam, which have no background category, changes in S_p of a particular ground object category will affect other ground object categories of interest. Since the SAM-generated masks originally contain overlapping regions, when S_n and $S_{n'}$, which have overlapping regions, are selected by different categories of S_p, and when the RS object S_p is replaced by the SAM object S_n with a larger area that overlaps with different categories of $S_{p'}$, ambiguous labeling regions are generated during the restoration of the full semantic segmentation image, as shown Fig. 4(c). We compute the overlapping regions of different categories of objects based on the enhanced RS object list P^S and convert these overlapping regions into the overlapping

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

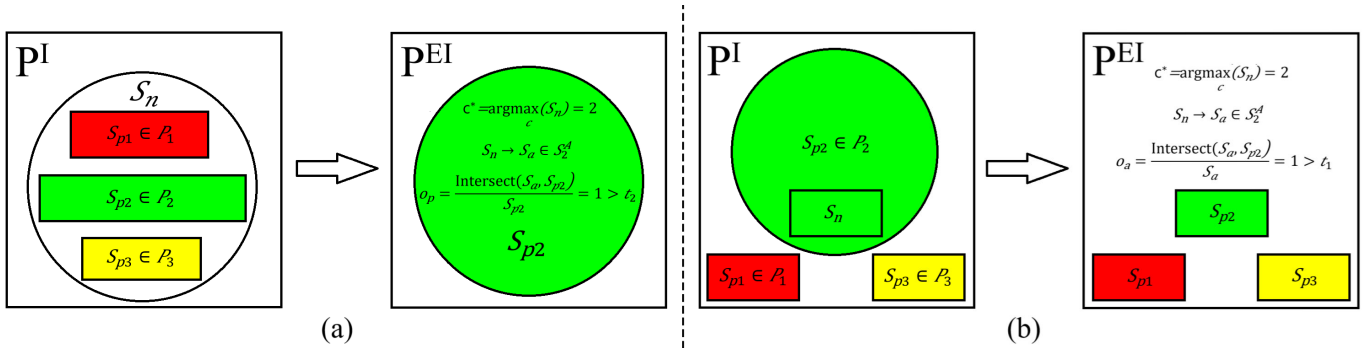


Fig. 3. Schematic of the error in enhance P^I to P^{EI}. (a) When area$(S_n) \gg$ area(S_p), as long as RS object S_p is the object of category c, which has the most pixels in S_n, then S_n is assigned to S_c^A. Due to the large difference in area, condition (3) is easily satisfied. Intuitively, the SAM recognizes different categories of ground objects as a whole. (b) When area$(S_p) \gg$ area(S_n), condition (2) is easily satisfied too, then S_n is assigned to S_c^A,c is the category of S_p. Intuitively, SAM detects small objects with distinctive features and fails to recognize them as a whole with more ambiguous boundaries. The superscript E indicates the enhanced result.

Algorithm 1 Object Assignment and Object Selection

Input: Remote Sensing Object Mask P, SAM Object Mask list S, threshold t_1^c, t_2^c
Output: Enhanced Remote Sensing Object Mask list P^S

procedure OAOS(P, S, t_1^c, t_2^c)
  $S^A = [S_1^A, \ldots, S_C^A]$ where $S_c^A = \{\}$
  $P^S = [P_1^S, \ldots, P_C^S]$ where $P_c^S = \{\}$
  for n from 0 to N

$$c^* = \frac{S_n\left(S_n == \arg\max(S_n)_c\right)}{S_n} > t_1^c$$

    $S_{c^*}^A \leftarrow S_{c^*}^A \cup \{S_n\}$
  for c from 0 to C
  if $P_c = \{\}$
    continue
  else
    for each mask S_p in P_c
    $tmp \leftarrow \{\}$
    for each mask S_a in S_c^A
    $o_a = \frac{\text{Intersect}(S_a, S_p)}{S_a}$
    $o_p = \frac{\text{Intersect}(S_a, S_p)}{S_p}$
    if $o_a > t_1^c$ **or** $o_p > t_2^c$
    $tmp \leftarrow tmp \cup \{S_a\}$
    if $tmp = \{\}$
    continue
    else if $number_{tmp} == 1$ and $t_2^c * S_p < tmp_1 < \frac{S_p}{t_1^c}$
    $S_p \leftarrow tmp_1$
    else if $\frac{\text{Intersect}(\text{Merge}(tmp), S_p)}{S_p} > t_2^c$
    $S_p \leftarrow \text{Merge}(tmp)$
    $P_c^S \leftarrow P_c^S \cup \{S_p\}$
end procedure

object list I, $I = [I_1, \ldots, I_N]$, where N is the number of overlapping objects. These objects are then sorted by iou to determine the assignment order for each category c, creating the overlapping assignment object list I^A, $I^A = [I_1^A, \ldots, I_C^A]$ which resolves ambiguous labeling issues.

In addition, since the SAM-generated masks sometimes contain localized missing regions, and the semantic mask S_p may be reduced in area after being replaced by the SAM mask S_n, restoring the full semantic segmentation image can result in partially missing masks, as shown Fig. 4(d). Based on the semantic segmentation image P^I, we determine the category c_b that has the most pixels at the edge of each RS object and replace the original category c of the RS object with c_b. This yields the semantic segmentation base image P^{BI}, which is used to fill the regions with missing masks after restoration.

Finally, we convert the SAM-enhanced object list P^{SM} into an image P^{SMI}, the overlapping object assignment list I^A into an image I^{AI}, and merge P^{BI}, P^{SMI}, and I^{AI} based on their order to obtain the final enhanced semantic segmentation image P^{EI}, as shown in Fig. 1(c).

The algorithms mentioned in this section are summarized in Algorithms 2.

IV. Experiments

A. Datasets

1) ISPRS Potsdam: The Potsdam dataset contains 38 high-resolution images of 6000 × 6000 pixels covering the city of Potsdam, Germany, with a spatial resolution of 5 cm. The dataset provides two types of annotations with non-eroded (NE) and eroded (E) options, which are distinguished as bounded and unbounded, respectively. In this work, we use TOP images from the Potsdam dataset and exclude DSM images. Since our framework uses SAM to enhance prediction boundaries and improve performance, we use the NE label for benchmarking, unlike most papers that use the eroded (E) label [6], [7], [23]. Following the experimental setup [6], we divide the dataset into 24 images for training and 14 images for testing. The test set includes images with index numbers 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13. This dataset includes five foreground classes, namely, *impervious surface*, *building*, *low vegetation*, *tree*, *car*, and one background class (*clutter*).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

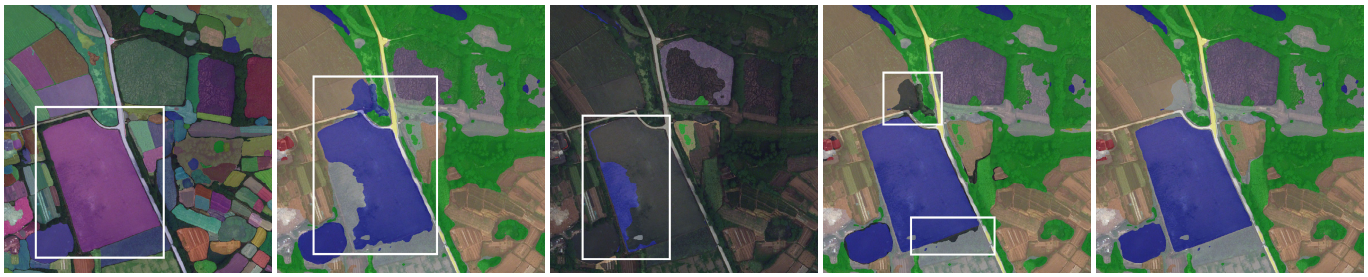


Fig. 4. When the SAM object in the framed area of (a) is matched with the RS object in the framed area of (b), the SAM mask replaces the corresponding area of the semantic mask. This creates an overlapping area in the framed area of (c) (overlap of water and background) and results in a missing area in the framed area of (d). (e) Desired outcome.

Algorithm 2 Restore Remote Sensing Semantic Segmentation Image

Input: Remote Sensing Object Mask list P, Enhanced Remote Sensing Object Mask list P^S
Output: Enhanced Remote Sensing Semantic Segmentation Image P_{EI}

procedure RESTORE(P, P^S)
  $P_c^{SM} \leftarrow$ Merge all objects in P^S into one object
  $I = Intersect_{c1,c2 \in C, c1 \neq c2}(P_{c1}^{SM}, P_{c2}^{SM})$

  $I^A \leftarrow [I_1^A, \ldots, I_C^A]$ where $I_c^A = \{\}$
  for each mask S_i in I
  $S_p^* = \arg\max_{S_p \in P} iou(S_i, S_p)$
  $c^* \leftarrow$ the category of S_p^*
  $I_{c^*}^A \leftarrow I_{c^*}^A \cup S_i$

  $P^B \leftarrow [P_1^B, \ldots, P_C^B]$ where $P_c^B = \{\}$
  for each mask S_p in P_c
  $P_s \leftarrow$ surrounding pixels of S_p
  $c_b^* = \arg\max_c P_s$
  $P_{c_b^*}^B \leftarrow P_{c_b^*}^B \cup S_p$

  $P^{BI}, P^{SMI}, I^{AI} \leftarrow$ Converts each object list P^B, P^{SM}, I^A to an image
  $P^{EI} \leftarrow$ Overlay P^{BI}, P^{SMI}, I^{AI} from top to bottom
end procedure

Each image is overlapped and divided into subimages of size 1024 × 1024 with a step size of 512 × 512.

2) LoveDA Urban: The LoveDA dataset [47]contains 5987 images with a resolution of 1024 × 1024 pixels and a spatial resolution of 30 cm. The images capture diverse urban scenes from three cities in China (Nanjing, Changzhou, and Wuhan) and include 166 768 annotated objects. The dataset includes seven land-cover classes: *background*, *building*, *road*, *water*, *barren*, *forest*, and *agriculture*. LoveDA is divided into two domains: Urban and Rural. For this study, we focus on the Urban domain, consistent with the experimental setup described in [23]. In LoveDA Urban, 1833 images are used, with 1156 images for training and 677 for testing. The training set consists of images indexed from 1366 to 2521, while the test set spans from 3514 to 4190.

Comparing the two datasets, although both are high-resolution remote sensing datasets, ISPRS Potsdam has a higher spatial resolution, fewer ground object categories, greater labeling accuracy, and a larger number of samples. During the review of the label quality in the LoveDA Urban dataset, we observed that most samples were labeled using simple polygon-based techniques, which resulted in discrepancies between the labeled boundaries and the actual feature boundaries, as shown in Fig. 5. When SAM generates masks with more accurate feature boundaries, this mismatch in label boundaries significantly impacts the evaluation of our proposed method. The discrepancies arise from the limited prior knowledge of the labelers, leading to slight differences in the labeling of the same ground object by different individuals [48]. However, water and buildings are generally ground object categories with less labeling ambiguity. To address this, we used the Label Studio tool to relabel all the instances of water and buildings in the 677 validation samples of the LoveDA Urban dataset, while leaving other ground object categories unchanged. The modified dataset is referred to as LoveDA Urban_m, and the corresponding truth values are termed the modified ground truth.

B. Evaluation Metrics

To evaluate the performance of the framework, we use four commonly used metrics: precision, recall, intersection over union (IoU), and $F1$ score ($F1$). Recall focuses on how many true positives were captured by the model, and precision focuses on the accuracy of those predictions. IoU and $F1$ can be used to weigh precision and recall, which can effectively avoid the short-board effect and effectively deal with the category imbalance problem so as to provide a more comprehensive assessment. These widely adopted statistical indices allow for a fair comparison of our method's performance with state-of-the-art baseline methods. The recall, precision, IoU, and $F1$ for category c are calculated using the following equations:

$$\text{recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{8}$$

$$\text{precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

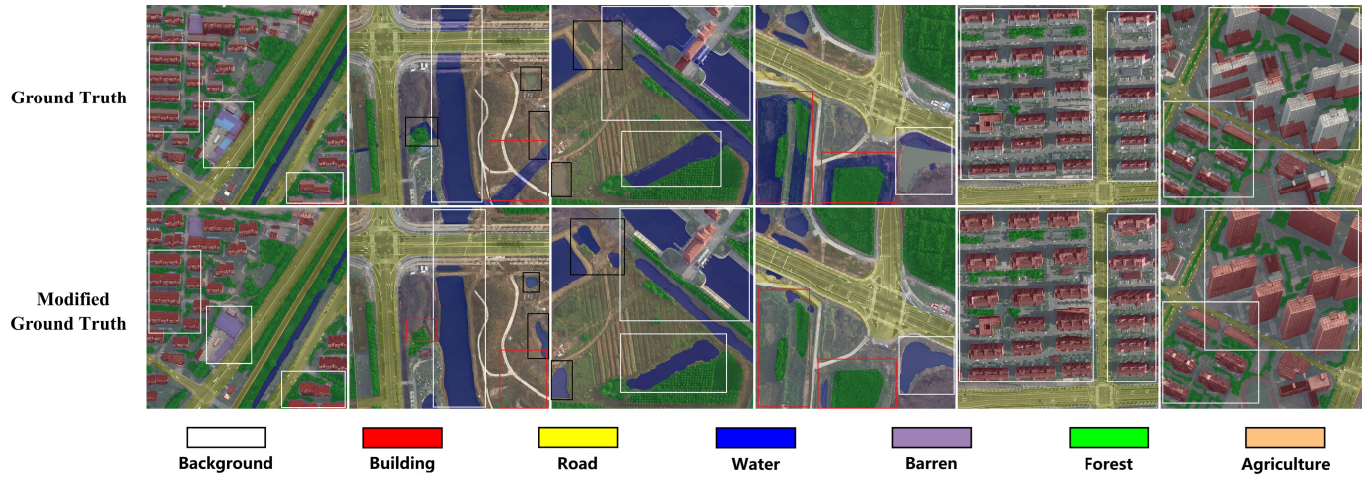


Fig. 5. Comparative visualization of the validation set in the LoveDA Urban dataset before and after modification. The top row displays the original ground truth, while the bottom row shows the modified ground truth. The white marking box highlights areas where the modified label contours are more consistent with the actual ground object boundaries. The red marking box indicates the correction of mislabeling present in the original labels, and the black marking box demonstrates the resolution of label omissions in the original dataset. These modifications contribute to improving the accuracy of the annotations in LoveDA Urban_m.

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{10}$$

$$F1_c = \frac{2\mathrm{TP}_c}{2\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c} \tag{11}$$

where TP_c, TN_c, FP_c, and FN_c represent the true positives, true negatives, false positives, and false negatives for class c, respectively. After calculating $F1$ and IoU for the main classes as defined above, we can derive their averages, denoted as $mF1$ and mIoU.

C. Implementation Details

The experiments were conducted using PyTorch on a NVIDIA RTX A5000 GPU equipped with 24-GB RAM. To generate SAM objects, we used the interface provided by Meta AI, which involves five relevant parameters: "crop_nms_thresh," "box_nms_thresh," "stability_score_thresh," "pred_iou_thresh," and "min_region." The definitions of these parameters can be found in the Meta AI documentation. In our experiments, these values were set to 0.5, 0.5, 0.95, 0.95, and 500, respectively. Note that higher values of "pred_iou_thresh" and "stability_score_thresh" tend to produce more reliable raw outputs. Similarly, larger values of "crop_nms_thresh" and "box_nms_thresh" will reduce redundant outputs, while larger values of "min_region" will filter out smaller masks that lack clear meaning. In addition, since the focus of this work is on assessing whether SAM can serve as a postprocessing method to improve the accuracy of semantic segmentation in remote sensing, rather than evaluating the individual models, the hyperparameters affecting model performance were not extensively fine-tuned. The initial learning rate for all the models was set to 1e-4, with the initial learning rate for the backbone network set to 1e-5. The weight decay coefficient was configured at 5e-4, and the batch size was set to 8. The maximum number of epochs was 200, with early stopping applied if accuracy did not improve for 20 consecutive epochs. Adam with Weight Decay (AdamW) was used as the optimization algorithm for training all the models. A combination of soft cross-entropy loss and Dice loss, fused at a 1:1 ratio, was selected for the loss function. Simple data augmentations, such as random rotations and flips, were applied in all the experiments. For the parameters t_1^c and t_2^c used in our method, the optimal values were determined through a grid search on the validation set of the dataset, as analyzed in Section IV-E.

D. Performance Comparison

The existing semantic segmentation models can be classified into four main categories based on the type of encoder and decoder [7]: 1) CNN-based lightweight networks developed for efficient semantic segmentation; 2) CNN-based attentional networks; 3) hybrid transformer-based networks, which feature a transformer-based encoder and a CNN-based decoder; and 4) fully transformer-based networks with both a transformer-based encoder and decoder. For each of these four categories, we selected two representative models: ABCNet [6], ShelfNet [34], FANet [35], MANet [36], DC-Swin [39], UNetformer [7], SegFormer [37], and FT-UNetformer [7], to test our proposed framework. This selection allows us to validate our method across various well-established networks. The quantitative results are presented in Tables I–III.

1) Performance Comparison on the LoveDA Urban Dataset: As shown in Table I, our proposed framework demonstrates significant improvements in both mIoU and mF1 metrics compared with the eight baseline methods. These results confirm the stability and effectiveness of our framework in enhancing semantic segmentation for remote sensing. Notably, unlike other methods, our framework requires no additional training. It improves segmentation performance by optimizing the two best parameters, t_1^c and t_2^c, on the validation set, retaining the mask that best matches the dataset in the SAM mask list, and using it to refine the boundaries of the segmented RS semantic objects. To illustrate this, we use ABCNet as an example. ABCNet + SAM, ABCNet + HQ-SAM, and ABCNet + Semantic-SAM outperform ABCNet across all seven ground objects categories (*background*, *building*, *road*, *water*, *barren*, *forest*, and *agricultural*). Notably, the eval-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EXPERIMENTAL RESULTS ON THE LOVEDA URBAN DATASET. THE ACCURACY OF EACH CATEGORY IS PRESENTED IN THE IOU/F1 FORM. BOLD VALUES ARE THE BEST

Method	Backbone	background	building	road	water	barren	forest	agricultural	mIoU	mF1
ABCNet	ResNet-18	42.63/59.78	60.76/75.59	56.13/71.9	66.6/79.95	35.27/52.14	41.45/58.61	49.79/66.48	50.38	66.35
ABCNet+SAM	ResNet-18	42.86/60.0	60.92/75.71	57.57/73.07	67.78/80.8	36.76/53.76	41.62/58.78	50.32/66.95	51.12	67.01
ABCNet+HQ-SAM	ResNet-18	42.79/59.93	**60.99/75.77**	57.42/72.95	67.75/80.78	36.86/53.87	41.74/58.9	50.6/67.2	51.16	67.06
ABCNet+Semantic-SAM	ResNet-18	**42.88/60.03**	60.96/75.74	**57.85/73.29**	**67.83/80.83**	**37.28/54.31**	**41.77/58.93**	**50.84/67.41**	**51.34**	**67.22**
ShelfNet	ResNet-18	41.94/59.09	61.67/76.29	57.48/73.0	68.37/81.22	14.21/24.89	42.56/59.71	47.2/64.13	47.63	62.62
ShelfNet+SAM	ResNet-18	42.23/59.38	61.84/76.42	58.32/73.68	68.98/81.64	16.14/27.79	42.73/59.87	48.01/64.87	48.32	63.38
ShelfNet+HQ-SAM	ResNet-18	42.27/59.43	61.86/76.44	58.31/73.67	**69.07/81.71**	16.13/27.78	42.82/59.96	48.19/65.04	48.38	63.43
ShelfNet+Semantic-SAM	ResNet-18	**42.39/59.54**	**62.01/76.55**	**58.57/73.87**	69.02/81.67	**17.04/29.11**	**43.02/60.16**	**49.0/65.77**	**48.72**	**63.81**
FANet	ResNet-18	36.15/53.1	52.36/68.73	51.79/68.24	63.07/77.35	31.22/47.59	36.85/53.86	6.19/11.66	39.66	54.36
FANet+SAM	ResNet-18	**36.34/53.31**	52.47/68.82	52.29/68.67	64.27/78.25	31.43/47.82	37.06/54.08	6.27/11.8	40.02	54.68
FANet+HQ-SAM	ResNet-18	36.32/53.29	52.46/68.82	**52.32/68.7**	**64.41/78.35**	**31.59/48.01**	37.05/54.06	6.33/11.9	**40.07**	**54.73**
FANet+Semantic-SAM	ResNet-18	36.31/53.28	**52.52/68.87**	52.12/68.53	64.35/78.31	31.43/47.82	**37.08/54.1**	**6.4/12.02**	40.03	54.7
MANet	ResNet-18	38.56/55.66	59.06/74.27	52.58/68.92	67.88/80.87	35.63/52.54	37.38/54.42	36.02/52.97	46.73	62.81
MANet+SAM	ResNet-18	38.48/55.58	59.21/74.38	53.12/69.38	68.01/80.96	36.95/53.96	37.77/54.83	36.77/53.77	47.19	63.27
MANet+HQ-SAM	ResNet-18	38.75/55.86	59.23/74.39	**53.65/69.83**	68.17/81.08	38.3/55.38	37.83/54.89	37.41/54.46	47.62	63.7
MANet+Semantic-SAM	ResNet-18	**38.86/55.97**	**59.42/74.54**	53.54/69.74	**68.27/81.15**	**38.8/55.9**	**38.34/55.43**	**39.01/56.12**	**48.03**	**64.12**
DC-Swin	Swin-Base	42.3/59.45	63.89/77.97	64.32/78.28	72.0/83.72	33.46/50.14	43.68/60.8	48.05/64.91	52.53	67.9
DC-Swin+SAM	Swin-Base	42.43/59.58	63.99/78.04	64.41/78.35	72.01/83.73	34.71/51.54	43.83/60.94	48.4/65.23	52.83	68.2
DC-Swin+HQ-SAM	Swin-Base	42.41/59.56	63.96/78.02	**64.45/78.38**	**72.05/83.76**	34.72/51.55	43.84/60.95	48.64/65.45	52.87	68.24
DC-Swin+Semantic-SAM	Swin-Base	**42.51/59.66**	**64.05/78.08**	64.42/78.36	72.04/83.75	**35.26/52.14**	**44.05/61.16**	**49.01/65.78**	**53.05**	**68.42**
UNetFormer	ResNet-18	42.81/59.96	59.02/74.23	52.92/69.21	69.25/81.83	38.25/55.33	41.03/58.18	45.11/62.18	49.77	65.85
UNetFormer+SAM	ResNet-18	43.04/60.18	59.25/74.41	53.72/69.89	69.51/82.01	39.38/56.51	41.19/58.35	45.58/62.62	50.24	66.28
UNetFormer+HQ-SAM	ResNet-18	43.03/60.17	59.26/74.42	53.66/69.84	**69.63/82.1**	39.56/56.7	41.25/58.41	45.73/62.76	50.3	66.34
UNetFormer+Semantic-SAM	ResNet-18	**43.15/60.29**	**59.36/74.5**	**53.92/70.06**	69.62/82.09	**39.91/57.05**	**41.49/58.65**	**46.46/63.45**	**50.56**	**66.58**
SegFormer	MiT-B3	42.68/59.83	65.38/79.07	61.7/76.31	72.6/84.12	28.3/44.11	45.07/62.14	51.25/67.76	52.42	67.62
SegFormer+SAM	MiT-B3	42.81/59.95	65.42/79.1	**62.12/76.64**	72.67/84.17	29.19/45.18	45.22/62.27	51.75/68.2	52.74	67.93
SegFormer+HQ-SAM	MiT-B3	42.77/59.91	65.42/79.09	62.04/76.58	**72.75/84.23**	29.33/45.36	45.16/62.22	51.77/68.22	52.75	67.94
SegFormer+Semantic-SAM	MiT-B3	**42.97/60.11**	**65.44/79.11**	61.94/76.49	72.74/84.22	**30.57/46.82**	**45.35/62.4**	**52.4/68.76**	**53.06**	**68.27**
FT-UNetFormer	Swin-Base	38.07/55.14	61.67/76.29	60.16/75.12	66.48/79.87	36.51/53.49	38.76/55.87	42.5/59.65	49.16	65.06
FT-UNetFormer+SAM	Swin-Base	38.27/55.36	61.78/76.38	60.29/75.23	67.68/80.73	38.06/55.14	39.02/56.13	43.06/60.19	49.74	65.59
FT-UNetFormer+HQ-SAM	Swin-Base	38.24/55.32	61.83/76.41	**60.39/75.31**	**67.79/80.8**	38.05/55.12	39.07/56.19	43.16/60.29	49.79	65.64
FT-UNetFormer+Semantic-SAM	Swin-Base	**38.31/55.39**	**61.96/76.51**	60.37/75.29	67.54/80.62	**38.5/55.6**	**39.1/56.22**	**43.4/60.53**	**49.88**	**65.74**

uation metrics for the *barren*, *road*, and *water* categories increase by more than 1%. Furthermore, all the semantic segmentation models postprocessed with the visual foundation model show improvements of 0.6% to 1.3% in the integrated evaluation metrics, mIoU and mF1. The Semantic-SAM model demonstrates greater improvement compared with SAM-HQ and SAM, suggesting that the potential of our framework to enhance semantic segmentation performance in remote sensing will continue to grow with further development of the visual foundation model.

Fig. 6 provides an example visualization of LoveDA Urban, with white and black marking boxes highlighting regions of interest across all the subgraphs. Our framework enhances the model's ability to more accurately recognize barren, as demonstrated in all the model experiments. The original semantic segmentation model achieves high accuracy but suffers from low recall when recognizing barren, whereas SAM provides precise barren boundaries. This serves as an effective solution to the underprediction issue mentioned in this article through our algorithm. In addition, our method clearly improves the semantic segmentation model's recognition of *water*, *agricultural*, and wider *road*. However, for the *forest* category, the semantic segmentation model predicts with low accuracy, and SAM's performance in recognizing forest boundaries is also limited due to the indistinct characteristics of *forest* objects, resulting in only a minor improvement when our method is applied. It is also worth noting that although *building* has distinct object characteristics and SAM achieves high accuracy in boundary recognition, the improvement in accuracy using our method remains minimal—an outcome that does not meet expectations. The LoveDA Urban dataset generally provides correct labeling for most ground objects, but its precision in outlining ground object boundaries is limited. In addition, due to the influence of labelers' prior knowledge, slight variations may occur when different labelers annotate the same ground object [48]. However, *water* and *building* generally exhibit low labeling ambiguity. Therefore, we used the Label Studio tool to relabel all the *water* and *building* categories in the 677 validation images of the LoveDA Urban dataset.

We recalculated the evaluation metrics for each category using the modified labels, as shown in Table II. For illustration, ABCNet is taken as an example. Although ABCNet^{m} was not enhanced with SAM and only replaced the original ground truth with modified ground truth for validation, the accuracy metrics for both the *building* and *water* categories are higher than those of ABCNet. This indicates that despite the limitations of the original labels, the semantic segmentation model still learns knowledge closer to the real ground truth than to the labels. After being enhanced by SAM, ABCNet^{m} + SAM improves the accuracy metrics for *building* and *water* categories by about 3 points compared with ABCNet^{m}, demonstrating the strong postprocessing enhancement capability of our framework for ground objects with distinct object significance. In addition, ABCNet^{m} + SAM improves the overall evaluation metrics, mIoU and mF1, by approximately 1.5 points over ABCNet^{m}, which is also a greater improvement than that of ABCNet + SAM over ABCNet. Our method not only improves semantic segmentation results in evaluation metrics but also produces enhanced predictions that are closer to the real-world conditions, particularly when validation labeling accuracy is limited.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

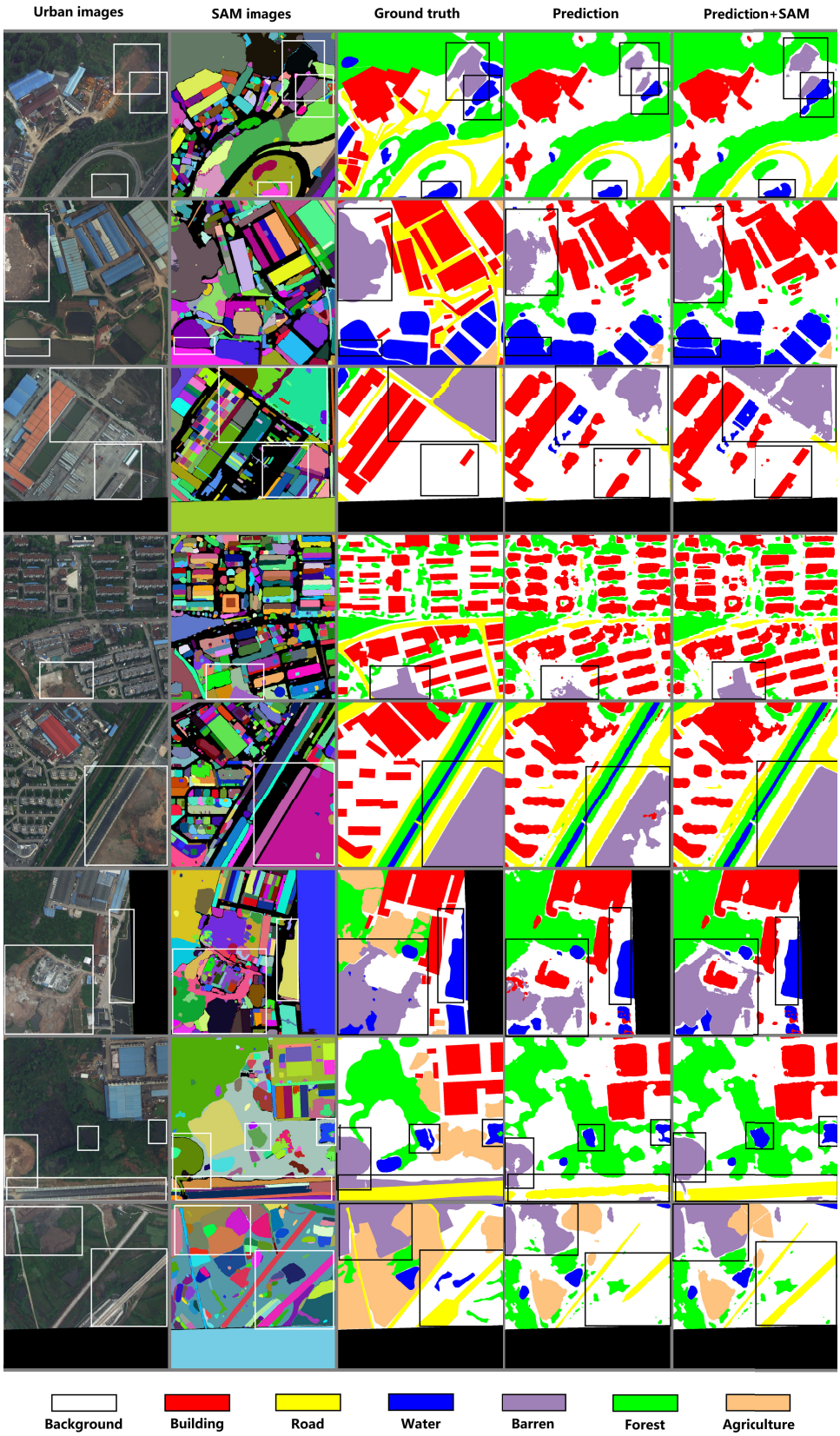


Fig. 6. Qualitative performance comparisons on the LoveDA Urban with the size of 1024 × 1024. For each of the four types of networks, we selected two representative models. The semantic segmentation models, listed from top to bottom, are ABCNet, ShelfNet, FANet, MANet, DC-Swin, UNetformer, SegFormer, and FT-UNetformer. Some white and black marking boxes are added to highlight the differences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

TABLE II

EXPERIMENTAL RESULTS ON THE LOVEDA URBAN DATASET. THE ACCURACY OF EACH CATEGORY IS PRESENTED IN THE IOU/F1 FORM. THE SUPERSCRIPT M REPRESENTS THE VALIDATION SET AS THE MODIFIED LOVEDA URBAN. BOLD VALUES ARE THE BEST

Method	Backbone	background	building	road	water	barren	forest	agricultural	mIoU	mF1
ABCNet	ResNet-18	42.63/59.78	60.76/75.59	56.13/71.9	66.6/79.95	35.27/52.14	41.45/58.61	49.79/66.48	50.38	66.35
ABCNet+SAM	ResNet-18	42.86/60.0	60.92/75.71	57.57/73.07	67.78/80.8	36.76/53.76	41.62/58.78	50.32/66.95	51.12	67.01
ABCNet^{m}	ResNet-18	42.84/59.98	66.04/79.55	56.22/71.98	76.72/86.83	35.32/52.21	41.58/58.74	50.38/67.0	52.73	68.04
ABCNet^{m}+SAM	ResNet-18	**43.6/60.72**	**69.3/81.87**	**57.61/73.11**	**79.17/88.38**	**36.79/53.79**	**41.76/58.92**	**50.91/67.47**	**54.16**	**69.18**
MANet	ResNet-18	38.56/55.66	59.06/74.27	52.58/68.92	67.88/80.87	35.63/52.54	37.38/54.42	36.02/52.97	46.73	62.81
MANet+SAM	ResNet-18	38.48/55.58	59.21/74.38	53.12/69.38	68.01/80.96	36.95/53.96	37.77/54.83	36.77/53.77	47.19	63.27
MANet^{m}	ResNet-18	40.17/57.32	68.42/81.25	52.65/68.98	80.48/89.18	35.75/52.67	37.71/54.77	36.04/52.99	50.18	65.31
MANet^{m}+SAM	ResNet-18	**40.83/57.99**	**71.09/83.1**	**53.2/69.45**	**81.67/89.91**	**37.07/54.09**	**38.06/55.14**	**36.8/53.81**	**51.25**	**66.21**
UNetFormer	ResNet-18	42.81/59.96	59.02/74.23	52.92/69.21	69.25/81.83	38.25/55.33	41.03/58.18	45.11/62.18	49.77	65.85
UNetFormer+SAM	ResNet-18	43.04/60.18	59.25/74.41	53.72/69.89	69.51/82.01	39.38/56.51	41.19/58.35	45.58/62.62	50.24	66.28
$\text{UNetFormer}^{\text{m}}$	ResNet-18	43.65/60.77	65.41/79.09	53.0/69.28	81.84/90.01	38.35/55.44	41.33/58.49	45.6/62.64	52.74	67.96
$\text{UNetFormer}^{\text{m}}$+SAM	ResNet-18	**44.52/61.61**	**68.33/81.18**	**53.88/70.03**	**83.84/91.21**	**39.55/56.69**	**41.45/58.61**	**45.98/62.99**	**53.93**	**68.9**
FT-UNetFormer	Swin-Base	38.07/55.14	61.67/76.29	60.16/75.12	66.48/79.87	36.51/53.49	38.76/55.87	42.5/59.65	49.16	65.06
FT-UNetFormer+SAM	Swin-Base	38.27/55.36	61.78/76.38	60.29/75.23	67.68/80.73	38.06/55.14	39.02/56.13	43.06/60.19	49.74	65.59
FT-$\text{UNetFormer}^{\text{m}}$	Swin-Base	39.71/56.84	70.02/82.37	60.24/75.19	80.43/89.15	36.64/53.63	39.15/56.27	42.78/59.93	52.71	67.63
FT-$\text{UNetFormer}^{\text{m}}$+SAM	Swin-Base	**40.45/57.6**	**71.57/83.43**	**60.4/75.31**	**82.6/90.47**	**38.37/55.46**	**39.34/56.46**	**43.28/60.41**	**53.71**	**68.45**

TABLE III

EXPERIMENTAL RESULTS ON THE ISPRS POTSDAM DATASET. THE ACCURACY OF EACH CATEGORY IS PRESENTED IN THE IOU/F1 FORM. BOLD VALUES ARE THE BEST

Method	Backbone	ImSurf	Building	LowVeg	Tree	Car	Clutter	mIoU	mF1
ABCNet	ResNet-18	84.25/91.45	**91.62/95.63**	74.26/85.23	**76.41/86.63**	84.01/91.31	42.45/59.6	75.5	84.97
ABCNet+SAM	ResNet-18	84.29/91.47	91.61/95.62	74.3/85.25	76.36/86.59	**84.06/91.34**	42.91/60.06	75.59	85.06
ABCNet+HQ-SAM	ResNet-18	84.28/91.47	91.6/95.62	74.31/85.26	76.35/86.59	84.06/91.34	43.43/60.56	75.67	85.14
ABCNet+Semantic-SAM	ResNet-18	**84.33/91.5**	91.59/95.61	**74.37/85.3**	76.37/86.6	84.02/91.31	**44.23/61.33**	**75.82**	**85.28**
ShelfNet	ResNet-18	84.16/91.4	90.45/**94.99**	73.61/84.8	**75.43/85.99**	82.87/90.63	42.91/60.06	74.91	84.64
ShelfNet+SAM	ResNet-18	**84.24/91.45**	**90.46**/94.99	73.58/84.78	75.38/85.96	**83.49/91.0**	43.37/60.5	**75.09**	84.78
ShelfNet+HQ-SAM	ResNet-18	84.23/91.44	90.46/94.99	73.61/84.8	75.32/85.92	83.19/90.83	**43.65/60.77**	75.08	**84.79**
ShelfNet+Semantic-SAM	ResNet-18	84.18/91.41	90.45/94.98	**73.64/84.82**	75.39/85.97	83.11/90.77	43.35/60.48	75.02	84.74
FANet	ResNet-18	79.63/88.66	87.76/93.48	69.45/81.97	**68.78/81.5**	76.64/86.78	29.13/45.12	68.56	79.58
FANet+SAM	ResNet-18	**79.78/88.76**	87.76/93.48	69.47/81.99	68.71/81.46	**77.93/87.6**	29.87/46.0	**68.92**	79.88
FANet+HQ-SAM	ResNet-18	79.77/88.75	**87.79/93.5**	69.47/81.99	68.68/81.43	77.5/87.32	**30.17/46.36**	68.9	**79.89**
FANet+Semantic-SAM	ResNet-18	79.7/88.7	87.79/93.5	**69.52/82.02**	68.71/81.45	76.97/86.98	30.03/46.18	68.78	79.81
MANet	ResNet-18	81.22/89.64	88.19/93.72	72.0/83.72	**74.46/85.36**	83.02/90.72	34.1/50.85	72.16	82.34
MANet+SAM	ResNet-18	81.26/89.66	88.22/93.74	71.99/83.72	74.44/85.35	83.11/90.78	34.75/51.58	72.29	82.47
MANet+HQ-SAM	ResNet-18	**81.33/89.7**	**88.26/93.77**	72.0/83.72	74.34/85.28	83.19/90.82	**35.21/52.08**	**72.39**	**82.56**
MANet+Semantic-SAM	ResNet-18	81.26/89.66	88.13/93.69	**72.17/83.84**	74.26/85.23	**83.28/90.87**	35.13/51.99	72.37	82.55
DC-Swin	Swin-Base	84.83/91.79	**92.29/95.99**	74.91/85.65	**76.36/86.59**	84.02/91.32	40.86/58.02	75.55	84.89
DC-Swin+SAM	Swin-Base	**84.85/91.8**	92.29/95.99	**74.93/85.67**	76.34/86.58	**84.16/91.4**	40.99/58.15	**75.59**	**84.93**
DC-Swin+HQ-SAM	Swin-Base	84.84/91.8	92.29/95.99	74.92/85.66	76.35/86.59	84.03/91.32	**41.01/58.17**	75.57	84.92
DC-Swin+Semantic-SAM	Swin-Base	84.83/91.79	92.29/95.99	74.9/85.65	76.35/86.59	84.02/91.32	40.94/58.1	75.56	84.91
UNetFormer	ResNet-18	83.44/90.97	90.98/**95.28**	74.08/85.11	**75.32/85.92**	**83.58/91.06**	41.67/58.83	74.85	84.53
UNetFormer+SAM	ResNet-18	**83.49/91.0**	90.95/95.26	74.07/85.11	75.3/85.91	83.57/91.05	42.02/59.17	74.9	84.58
UNetFormer+HQ-SAM	ResNet-18	83.47/90.99	**90.99**/95.28	**74.1/85.12**	75.3/85.91	83.58/91.06	**42.33/59.48**	**74.96**	**84.64**
UNetFormer+Semantic-SAM	ResNet-18	83.44/90.97	90.98/95.28	74.09/85.12	75.31/85.92	83.57/91.05	41.81/58.97	74.87	84.55
SegFormer	MiT-B3	82.95/90.68	90.73/95.14	74.26/85.23	**75.93/86.32**	83.57/91.05	37.44/54.48	74.15	83.82
SegFormer+SAM	MiT-B3	**83.01/90.71**	90.73/95.14	74.25/85.22	75.91/86.3	**83.73/91.15**	37.83/54.89	74.24	83.9
SegFormer+HQ-SAM	MiT-B3	82.98/90.7	90.73/95.14	74.25/85.22	75.91/86.31	83.62/91.08	37.99/55.06	74.25	83.92
SegFormer+Semantic-SAM	MiT-B3	82.97/90.69	**90.7/95.15**	**74.3/85.25**	75.91/86.3	83.65/91.1	**38.09/55.16**	**74.28**	**83.94**
FT-UNetFormer	Swin-Base	84.99/91.89	92.01/95.84	74.87/**85.63**	**76.18/86.48**	84.94/91.86	39.81/56.95	75.47	84.77
FT-UNetFormer+SAM	Swin-Base	**85.01/91.9**	92.01/95.84	**74.88**/85.63	76.17/86.47	84.93/91.85	39.99/57.14	**75.5**	84.8
FT-UNetFormer+HQ-SAM	Swin-Base	85.0/91.89	**92.02/95.85**	74.81/85.59	76.12/86.44	**84.96/91.87**	**40.07/57.22**	75.5	**84.81**
FT-UNetFormer+Semantic-SAM	Swin-Base	84.99/91.89	92.01/95.84	74.84/85.61	76.16/86.47	84.95/91.86	39.9/57.04	75.48	84.78

As shown in Fig. 7, when comparing with our modified true values, we found that many of the buildings in the original ground truth were labeled as simple box shapes, which do not closely match the actual contours of the buildings. In addition, there were some errors in the original labels, which we corrected one by one to reevaluate our method. In Fig. 7, we use black marking and white boxes to mark only *building* and *water*. By observing the comparison, it is apparent that due to the limited accuracy of the original labels, the enhanced results may show a decrease in the evaluation metric rather than an increase when the original labels are used as the validation set. However, when we validate using more realistic labels—our modified labels—the evaluation metrics for *building* and *water* show significant improvement. This demonstrates that our framework is not merely focused on improving evaluation metrics, but aims to make the classification results more aligned with real-world conditions.

2) Performance Comparison on the Potsdam Dataset: As shown in Table III, our proposed framework demonstrates a consistent improvement in mIoU and mF1 compared with the method without SAM enhancement, affirming the generalization and stability of our approach across different datasets. However, it is worth noting that the improvements in evaluation metrics for mIoU and mF1 on the Potsdam dataset range

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

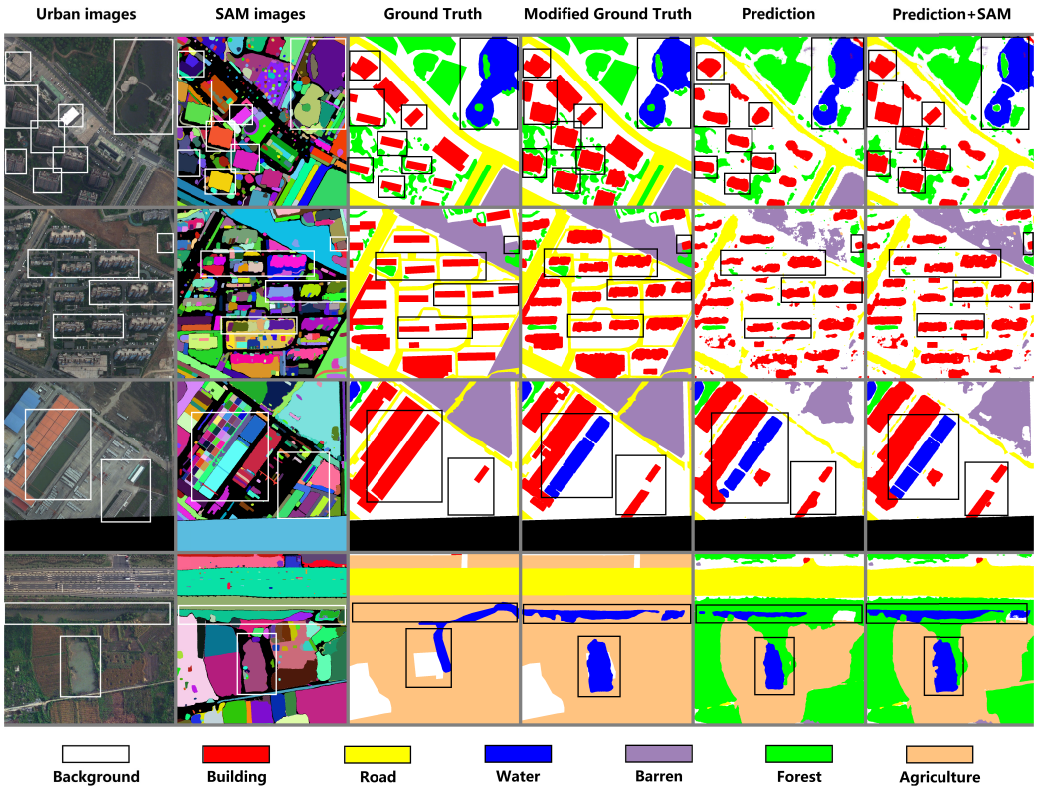


Fig. 7. Qualitative performance comparisons on the modified LoveDA Urban with the size of 1024 × 1024. For each of the four types of semantic segmentation models, we selected one representative model. The semantic segmentation models, listed from top to bottom, are ABCNet, ShelfNet, FANet, MANet, DC-Swin, UNetformer, SegFormer, and FT-UNetformer. Some white and black marking boxes are added to highlight the differences.

from 0.05% to 0.35%, which is significantly smaller than those observed on the LoveDA Urban dataset.

In several semantic segmentation model experiments, the accuracy for *building*, which has explicit object significance, not only fails to improve but also decreases, while the accuracy for *car*, which also has clear object meaning, shows little to no improvement. In contrast, the accuracy of *clutter*, which has the lowest baseline accuracy among the categories in the Potsdam dataset, shows the greatest improvement. A reasonable explanation is that our framework assumes that remote sensing semantic segmentation models cannot learn accurate ground object boundary information without sufficient and high-quality labels, whereas visual foundational models can provide more accurate ground object boundary information than remote sensing models. However, in the case of the Potsdam dataset, the labeling accuracy is very high and abundant. As a result, the semantic segmentation model trained on this dataset is able to recognize most ground objects boundaries as accurately as the visual foundational model. In such cases, applying our framework may introduce negative effects. Nevertheless, for ground objects with low recognition accuracy by the semantic segmentation model (e.g., *clutter*), our framework remains applicable.

Fig. 8 provides an example visualization of the Potsdam dataset, with black marking and white boxes highlighting regions of interest in all the subimages. A comparison between the SAM images without semantic information and the predicted results reveals that the boundary accuracy of most RS objects in the predictions is even higher than that of the corresponding objects in the SAM images. Exceptions include a few cases, such as the *car* and *clutter* RS objects. These findings are consistent with the metric changes reported in the analysis above.

E. Sensitivity Analysis

The analysis focuses on the sensitivity of the two hyperparameters, t_1^c and t_2^c. The parameter t_1^c indicates that at least t_1^c part of the SAM object is covered by the RS object of category c. Its main purpose is to correct the underprediction of the RS object by adjusting it to an SAM object with a larger area. On the other hand, t_2^c indicates that the RS object of category c is covered by the SAM object by at least part of t_2^c, primarily to address the overprediction of the RS object by adjusting it to an SAM object with a smaller area. These two parameters are crucial for filtering out SAM objects that are both high-quality and well-matched with RS objects.

Given the correlation between t_1^c and t_2^c, we conducted joint sensitivity experiments for each category using ABCNet on LoveDA Urban's modified validation set, with t_1^c ranging from 0.5 to 0.9 and t_2^c ranging from 0.55 to 0.95. Comparing the results in Figs. 9 and 10, it is evident that generally, a smaller t_1^c increases the recall but decreases the accuracy of the feature category. Conversely, a smaller t_2^c decreases recall but improves accuracy. For certain application scenarios, specific categories may require either high recall or high accuracy. Our approach can achieve this by dynamically adjusting t_1^c and t_2^c.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

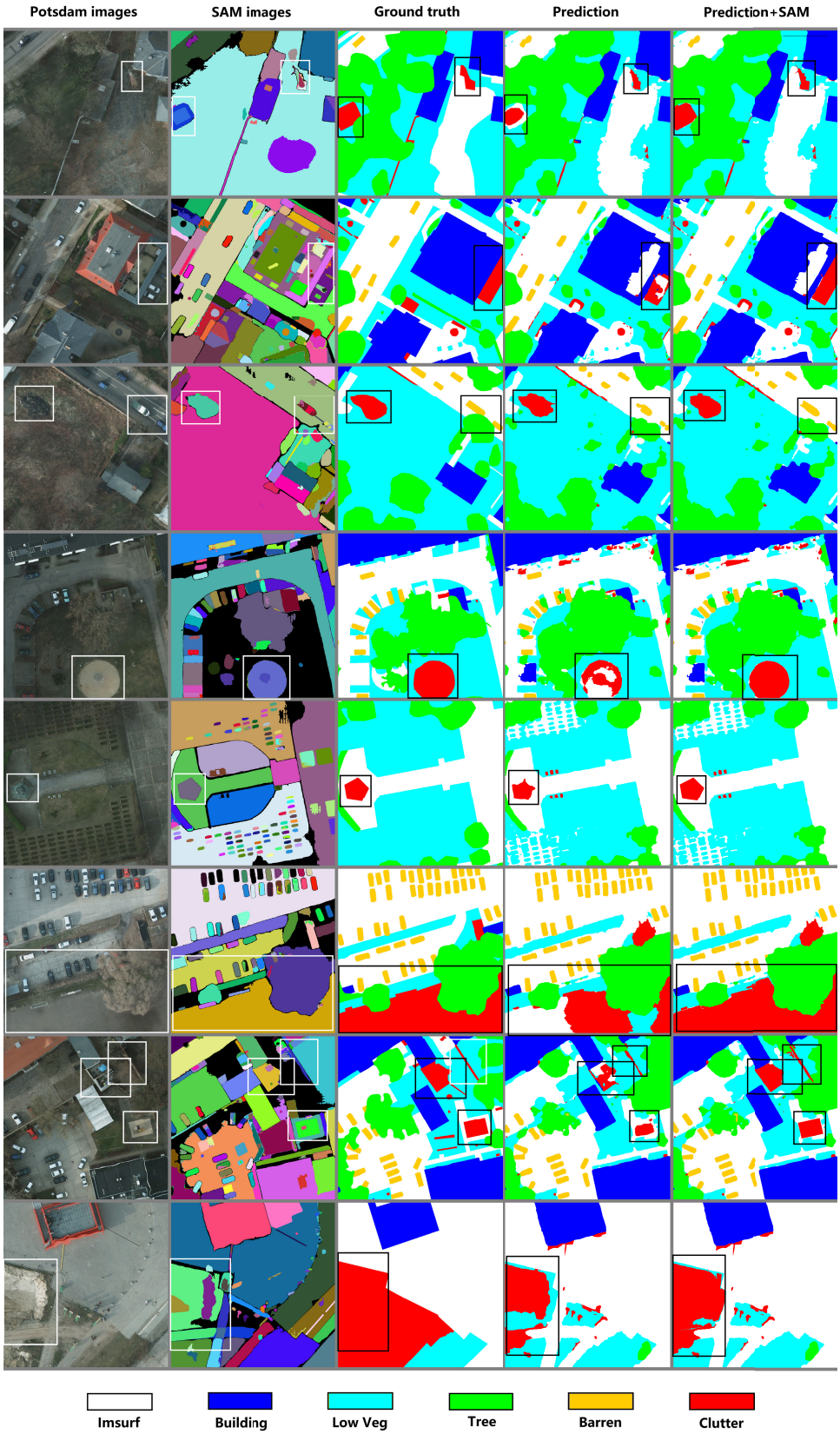


Fig. 8. Qualitative performance comparisons on the ISPRS Potsdam with the size of 1024 × 1024. For each of the four types of semantic segmentation models, we selected one representative model. The semantic segmentation models, listed from top to bottom, are ABCNet, ShelfNet, FANet, MANet, DC-Swin, UNetformer, SegFormer, and FT-UNetformer. Some white and black marking boxes are added to highlight the differences.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

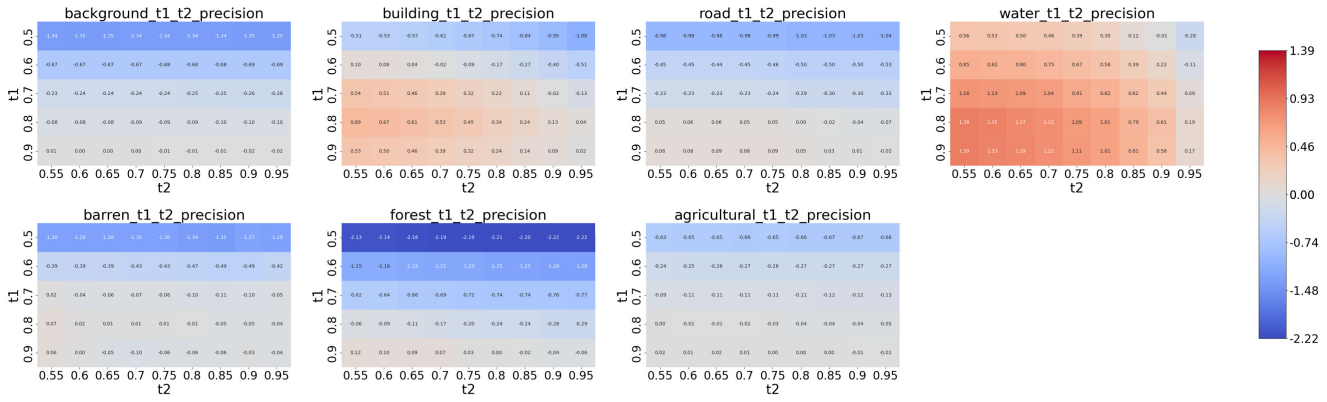


Fig. 9. Heat maps on changes in precision, with red representing an increase and blue a decrease.

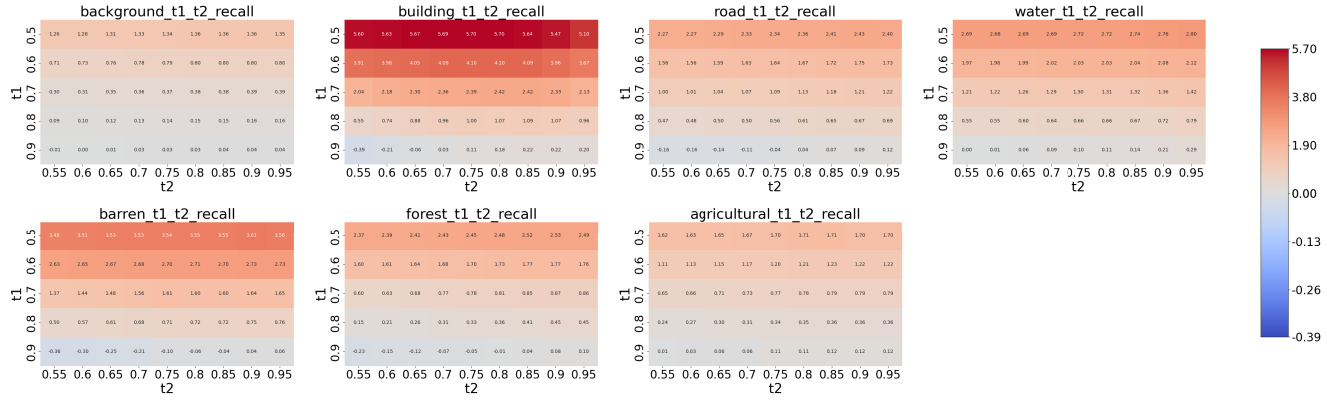


Fig. 10. Heat maps on changes in recall, with red representing an increase and blue a decrease.

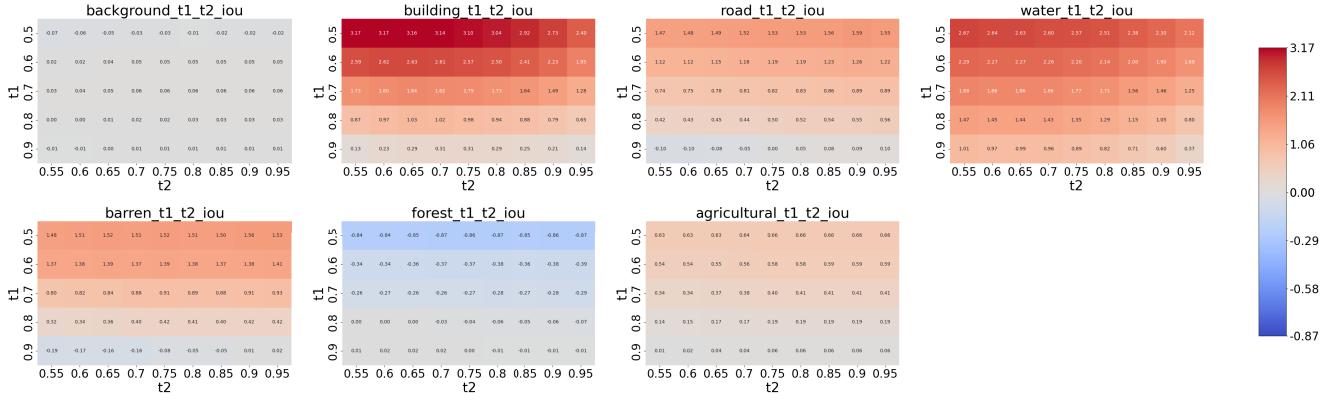


Fig. 11. Heat maps on changes in IoU, with red representing an increase and blue a decrease.

It is important to note that if t_1^c and t_2^c are not appropriately chosen for certain categories, it may result in performance degradation. Therefore, it is essential to traverse through different values to find the optimal t_1^c and t_2^c maximizing the effectiveness of SAM on RS objects. From the heat map variations in Fig. 11, it is clear that while t_1^c is tuned more finely than t_2^c, t_1^c remains the primary factor influencing the effectiveness of the enhancement.

Table IV shows the optimal values of t_1^c and t_2^c for each category, as identified by ABCNet on the validation set of the LoveDA Urban dataset. The evaluation metric, IoU, is effected by both recall and precision. Generally, recall and precision are negatively correlated, meaning that an increase in recall may result in a decrease in precision. When t_1^c is small, SAM primarily addresses the issue of underprediction by the semantic segmentation model, increasing recall but potentially reducing accuracy for categories such as *road*, *barren*, and *agriculture*. Conversely, when t_2^c is small, SAM primarily addresses the overprediction of categories, improving accuracy but potentially reducing recall for categories like *background* and *forest*. When both t_1^c and t_2^c are small, it indicates that the SAM-generated objects align well with this category, resulting

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

EXPERIMENTAL RESULTS ON THE MODIFIED LOVEDA URBAN DATASET. WE PRESENT THE OPTIMAL VALUES OF t_1^c AND t_2^c FOR THE SEVEN CLASSES AND THE CORRESPONDING CHANGES IN THEIR EVALUATION METRICS, WHERE THE SUPERSCRIPT E DENOTES THE ENHANCED RESULTS ACHIEVED BY OUR METHOD

Metric	background	building	road	water	barren	forest	agricultural
t_1^c	0.9	0.5	0.5	0.5	0.5	0.8	0.6
t_2^c	0.55	0.55	0.85	0.55	0.9	0.55	0.85
recall	59.81	82.02	61.21	83.86	46.29	73.07	64.69
recall^e	58.64	87.35	63.32	86.18	49.47	72.72	65.34
precision	60.15	77.22	87.34	90.01	59.85	49.1	69.48
precision^e	62.96	77.03	86.47	90.68	58.93	49.52	69.74
IoU	42.84	66.04	56.22	76.72	35.32	41.58	50.38
IoU^e	43.6	69.3	57.61	79.17	36.79	41.76	50.91

in the greatest improvement in evaluation metrics, as seen with *building* and *water*. When $t_1^c = 1.0$ and $t_2^c = 1.0$, it indicates that SAM does not enhance this category.

V. DISCUSSION AND CONCLUSION

A. Discussion

Although our approach, as a postprocessing method for semantic segmentation, demonstrates strong performance on two widely used datasets without requiring additional training, it still has some limitations that warrant further exploration. The performance of our method is primarily influenced by two factors: 1): whether SAM can generate accurate masks and 2): whether suitable hyperparameters, t_1^c and t_2^c, can be identified.

Regarding the first factor, previous research [20] has shown that SAM is sensitive to the resolution of remotely sensed imagery. As SAM was trained on very high-resolution natural images, it naturally performs better on high-resolution data. Current SAM design in remote sensing primarily focuses on images with a resolution below 0.5 m, which is also the case for our method. Whether SAM can be effectively applied to low- and medium-resolution remote sensing images remains an area for further investigation.

In addition, as shown in Fig. 11, it is evident that the performance of the original semantic segmentation model may degrade if t_1^c and t_2^c parameters are not properly selected. The values of t_1^c and t_2^c, presented in Table IV, represent the optimal hyperparameters we identified through validation set traversal. Although the primary goal of our work is to design a postprocessing method for supervised semantic segmentation using SAM to improve performance, when the validation set is unlabeled, we must rely on a priori knowledge to ensure our method functions correctly (e.g., when the original segmentation results have low recall, we reduce t_1^c; when the precision is low, we reduce t_2^c). The uncertainties in t_1^c and t_2^c limit the potential for extending our method to semi-supervised, weakly supervised, and other domains. Therefore, finding methods for adaptive parameter adjustment has become a key focus of our future work.

B. Conclusion

Recognizing that SAM can generate high-quality masks for certain remotely sensed features with clear boundaries, this study proposed a simple and generalized framework that fully incorporates the better boundary from the raw output of SAM to enhance remote sensing semantic segmentation models for better prediction. This novel approach performs postprocessing solely on the results of the original semantic segmentation, without requiring additional training or module redesign, making it both simple and efficient. Furthermore, we introduce two hyperparameters, t_1^c and t_2^c, to balance the underprediction and overprediction to reach the optimal segmentation for each category. By optimizing these two hyperparameters, the SAM mask that best matches the segmented remotely sensed image will be retained and thereby improve the prediction.

We validate the generality of our framework through extensive experiments on two widely used high-resolution datasets (with resolutions below 0.5 m) and four different types of semantic segmentation models, demonstrating the robustness of our approach. Comparing the validation experiments on two different datasets, an interesting phenomenon is reached that the proposed framework can better improve the performance of prediction for the one with poorer training samples. The reason is that better boundary from SAM can effectively decrease the uncertainty induced by the worse samples. Since most of the applications retrieve their training samples either from the existing datasets or from manually labeling and the training samples are subsequently not as good as the public datasets with strict quality controlling, the proposed framework can greatly improve the segmentation results by lowering the uncertainty of samples, which is practically useful for applications in the real world.

REFERENCES

[1] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[2] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, p. 111716, May 2020.

[3] K. Pang, L. Weng, Y. Zhang, J. Liu, H. Lin, and M. Xia, "SGBNet: An ultra light-weight network for real-time semantic segmentation of land cover," *Int. J. Remote Sens.*, vol. 43, nos. 15–16, pp. 5917–5939, Aug. 2022.

[4] A. Gupta, S. Watson, and H. Yin, "Deep learning-based aerial image segmentation with open data for disaster impact assessment," *Neurocomputing*, vol. 439, pp. 22–33, Jun. 2021.

[5] L. Huang, B. Jiang, S. Lv, Y. Liu, and Y. Fu, "Deep-learning-based semantic segmentation of remote sensing images: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8370–8396, 2024.

[6] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.

[7] L. Wang et al., "Unetformer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[8] J. E. Ball, D. T. Anderson, and C. S. Chan, "Comprehensive survey of deep learning in remote sensing: Theories, tools, and challenges for the community," *J. Appl. Remote Sens.*, vol. 11, no. 4, Sep. 2017, Art. no. 042609.

[9] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[10] A. Ramesh et al., "Zero-shot text-to-image generation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jul. 2021, pp. 8821–8831.

[11] C. Zhang et al., "A comprehensive survey on segment anything model for vision and beyond," 2023, *arXiv:2305.08196*.

[12] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[13] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[14] C. Zhang et al., "Faster segment anything: Towards lightweight SAM for mobile applications," 2023, *arXiv:2306.14289*.

[15] K. Lei et al., "Segment anything in high quality," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, Jan. 2023, pp. 1–11.

[16] F. Li et al., "Semantic-SAM: Segment and recognize anything at any granularity," 2023, *arXiv:2307.04767*.

[17] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[18] D. Wang, J. Zhang, B. Du, D. Tao, and L. Zhang, "SAMRS: Scaling-up remote sensing segmentation dataset with segment anything model," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2023, pp. 1–12.

[19] Z. Lu, Z. Xiao, J. Bai, Z. Xiong, and X. Wang, "Can SAM boost video super-resolution?" 2023, *arXiv:2305.06524*.

[20] S. Ren et al., "Segment anything, from space?" in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 8340–8350.

[21] S. Shankar, L. A. Stearns, and C. J. van der Veen, "Segment anything in glaciology: An initial study implementing the segment anything model (SAM)," Res. Square [Preprint], Version 1, Jun. 2023. [Online]. Available: https://doi.org/10.21203/rs.3.rs-3011246/v1

[22] C. Zhang et al., "Enhancing USDA NASS cropland data layer with segment anything model," in *Proc. 11th Int. Conf. Agro-Geoinform. (Agro-Geoinform.)*, Jul. 2023, pp. 1–5.

[23] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, "SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5636916.

[24] L. P. Osco et al., "The segment anything model (SAM) for remote sensing applications: From zero to one shot," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103540.

[25] X. Li, C. Wen, Y. Hu, and N. Zhou, "RS-CLIP: Zero shot remote sensing scene classification via contrastive vision-language supervision," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103497.

[26] X. Qi, Y. Wu, Y. Mao, W. Zhang, and Y. Zhang, "Self-guided few-shot semantic segmentation for remote sensing imagery based on large vision models," in *Proc. Int. Conf. Mach. Learn. Intell. Commun.* Cham, Switzerland: Springer, Jan. 2023, pp. 69–80.

[27] J. Zhang et al., "Text2Seg: Remote sensing image semantic segmentation via text-guided visual foundation models," 2023, *arXiv:2304.10597*.

[28] S. Julka and M. Granitzer, "Knowledge distillation with segment anything (SAM) model for planetary geological mapping," in *Proc. Int. Conf. Mach. Learn., Optim., Data Sci.* Cham, Switzerland: Springer, Jan. 2023, pp. 68–77.

[29] Z. Yan et al., "RingMo-SAM: A foundation model for segment anything in multimodal remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625716.

[30] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[31] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Munich, Germany. Cham, Switzerland: Springer, 2015, pp. 234–241.

[32] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 3640–3649.

[33] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[34] J. Zhuang, J. Yang, L. Gu, and N. Dvornek, "ShelfNet for fast semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 847–856.

[35] P. Hu et al., "Real-time semantic segmentation with fast attention," *IEEE Robot. Autom. Lett.*, vol. 6, no. 1, pp. 263–270, Jan. 2021.

[36] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[37] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12077–12090.

[38] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.

[39] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[40] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[41] J. Li, X. Huang, and J. Gong, "Deep neural network for remote-sensing image interpretation: Status and perspectives," *Nat. Sci. Rev.*, vol. 6, no. 6, pp. 1082–1086, Nov. 2019.

[42] D. C. Duro, S. E. Franklin, and M. G. Dubé, "A comparison of pixel-based and object-based image analysis with selected machine learning algorithms for the classification of agricultural landscapes using SPOT-5 HRG imagery," *Remote Sens. Environ.*, vol. 118, pp. 259–272, Mar. 2012.

[43] Y. Wang, J. Zhang, M. Kan, S. Shan, and X. Chen, "Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12275–12284.

[44] J. Xie, X. Hou, K. Ye, and L. Shen, "CLIMS: Cross language image matching for weakly supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4483–4492.

[45] T. Chen, Z. Mai, R. Li, and W.-L. Chao, "Segment anything model (SAM) enhanced pseudo labels for weakly supervised semantic segmentation," 2023, *arXiv:2305.05803*.

[46] W. Ji, J. Li, Q. Bi, T. Liu, W. Li, and L. Cheng, "Segment anything is not always perfect: An investigation of SAM on different real-world applications," 2023, *arXiv:2304.05750*.

[47] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[48] M. Rahaman, M. M. Hillas, J. Tuba, J. F. Ruma, N. Ahmed, and R. M. Rahman, "Effects of label noise on performance of remote sensing and deep learning-based water body segmentation models," *Cybern. Syst.*, vol. 53, no. 6, pp. 581–606, Aug. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:06:35 UTC from IEEE Xplore. Restrictions apply.